

CPEG460-VLSI-Project2-Proposal
Xor encryptor and decryptor
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Project Description:

The goal of the project is to design a circuit that takes user input and stores it as encrypted data. The data can later be retrieved in its encrypted form or decrypted to reveal the original contents, using 8 2-to-1 multiplexers to select between the two choices. This is accomplished by using an exclusive-or key chosen by the user, that is xor'ed with the users data and stored inside an 8 bit register for later retrieval.

I/O Table:

Type	Input Names							
Xor_key	k0	k1	k2	k3	k4	k5	k6	k7
User_Input	i0	i1	i2	i3	i4	i5	i6	i7
Selector	s							
clock	c							

Type	Output Names							
Mux_out	m0	m1	m2	m3	m4	m5	m6	m7

Section Descriptions:

Encryption(1):

Takes the input from the user as well as the xor encryption key and performs the encryption using the specific key. The result is stored in the next section in the register file. There are 8 2-input xor gates used in the combinational logic

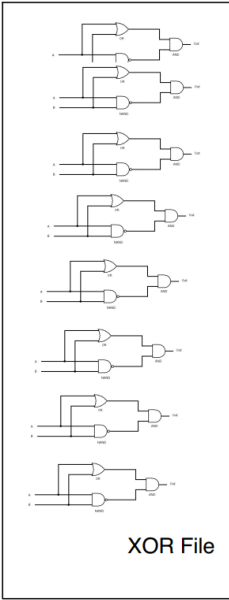
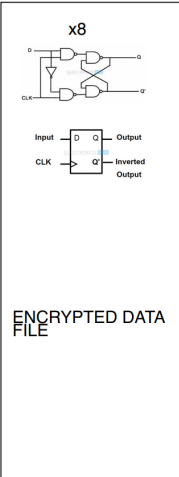
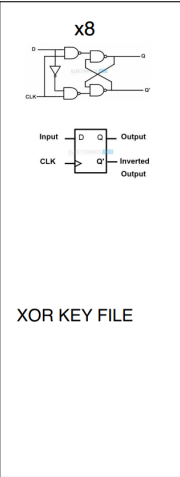
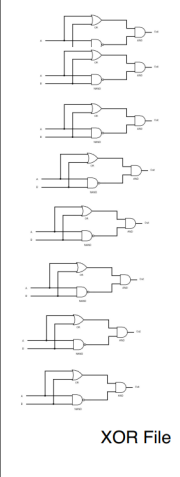
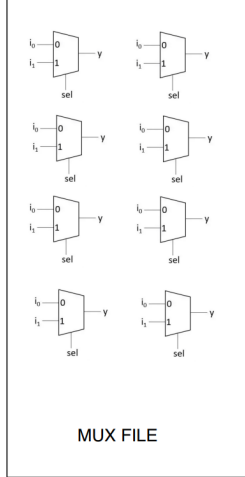
Storage(2):

The results of the previous stage encryption are stored here as well as the specific xor_key used. The results are stored in an 8 bit register file and the xor_key is stored in a separate 8 bit register file. D- flip flops are used in parallel to implement both files. The xor_key is stored because the user is allowed to change the key and the encryption used is kept consistent between clock cycles.

Decryption/output (3):

This stage takes its input from both register files. Both are passed into a second xor logic function to decrypt the stored data. The results that are output however are determined by the user. A mux-file takes both the encrypted and decrypted versions of the data and a selector is used to determine which output is displayed. The mux-file consists of 8 parallel 2-to-1 multiplexers.

Floor plan:

Encryption Section	Storage Section	Decryption/ display Section
 XOR File	 ENCIPHERED DATA FILE  XOR KEY FILE	 XOR File  MUX FILE
8 - parallel xor file Roughly (W x L) 20 x 104	2 - 8 bit register files side by side, utilizing D - flip flops Roughly(W x L) (20 x 104) each So (40 x 104)	1 duplicate xor file followed by an 8 2-to-1 mux file. The mux file is roughly (30 x 104) So together (50 x 104)

The length for each section is kept consistent for organization and the for the purpose of easy routing.